

Background: UX Evaluation

28

Highlights

- The dangers of trying to do summative evaluation in UX practice.
- Evaluation reliability.
- Variations in formative evaluation results.
- Roots for UX metrics and targets.
- The early A330 Airbus—An example of the need for ecological validity in testing.
- Determining the right number of participants.
- Roots of the critical incident data collection technique.
- More data collection techniques for emotional impact data.
- Nielsen and Molich’s original heuristics.
- UX problem data management.

28.1 THIS IS A REFERENCE CHAPTER

This chapter contains reference material relevant to the other chapters of [Part 5](#). This chapter is not intended to be read through as a regular chapter, but each section is supposed to be read when a reference to it in the main chapters is encountered.

28.2 THE DANGERS OF TRYING TO (OR EVEN APPEARING TO) DO FORMAL SUMMATIVE EVALUATION IN UX PRACTICE

28.2.1 Engineering Versus Science

It’s all very well in practice but it will never work in theory.

– French management saying

Sometimes, empirical lab-based UX testing that includes informal quantitative metrics is the source of controversy with respect to “validity.” Sometimes we hear,

Formal Summative Evaluation

Quantitative evaluation with a goal to assess, determine, or compare a level achieved of a parameter (such as usability), especially for assessing improvement in the user experience due to formative evaluation ([Section 21.1.5](#)).

Formal Summative Evaluation

A formal, statistically rigorous summative (quantitative) empirical UX evaluation that produces statistically significant results (Section 21.1.5.1).

“Because your informal summative evaluation was not controlled testing, why should we not dismiss your results as too ‘soft’?” “Your informal studies aren’t good science. You can’t draw any conclusions.”

These questions ignore the fundamental difference between formal and informal summative evaluation and the fact that they have completely different goals and methods. This may be due, in part, to the fact that the fields of HCI and UX were formed as a melting pot of people from widely varying backgrounds. From their own far-flung cultures in psychology, human factors engineering, systems engineering, software engineering, marketing, and management, they arrived at the docks of HCI with their baggage containing their own perspectives and mindsets.

Thus, it is known that formal summative evaluations are judged on a number of rigorous criteria, such as validity. But informal summative evaluation may be less known as an important engineering tool in the HCI bag, and that the *only* criterion for judging this kind of summative evaluation method is effectiveness within an *engineering* process.

28.2.2 What Happens in Engineering Stays in Engineering

Because informal summative evaluation is engineering, it comes with some very strict limitations, particularly on sharing informal summative results.

Informal summative evaluation results are only for internal use as engineering tools to do an engineering job by the project team and shouldn’t be shared outside the team. Because of the lack of statistical rigor, these results especially can’t be used to make any claims inside or outside the team. To make claims about UX levels achieved from informal summative results, for example, would be a violation of professional ethics.

We read of a case in which a high-level manager in a company got a UX report from a project team, but discounted the results because they were not statistically significant. This problem could have been avoided by following our simple rules and not distributing formative evaluation reports outside the team or by writing the report for a management audience with careful caveats.

Reliability, UX Evaluation

Refers to the repeatability of a UX method or technique from one UX practitioner to another and from one time to another for the same practitioner (Section 21.2.5.2).

28.3 UX EVALUATION RELIABILITY

The reliability of a UX evaluation method is about repeatability—how consistently can it be used to find the same results (UX problems to be fixed) across different evaluators and different designs being evaluated?

28.3.1 Individual Differences Naturally Cause Variations in Results

The literature contains many discussions and debates about the effectiveness and reliability of various UX evaluation methods. Effectiveness in the entire field of UX evaluation has been called into question. We can't hope to replicate that level of discussion here, but we do offer a few insights into the problem.

When the variation of UX evaluation results is due to using different evaluators, it is called the “evaluator effect” (Hertzum and Jacobsen, 2003; Vermeeren, van Kesteren, and Bekker, 2003). Different people see usage and problems differently and have different problem detection rates. They naturally see different UX problems in the same design. Different evaluators even report very different problems when observing the same evaluation session. Different UX teams interpret the same evaluation report in different ways. Even the same person can get different results in two successive evaluations of the same system.

28.3.2 Why So Much Variation? UX Evaluation is Difficult

In our humble opinion, the biggest reason for the limitations of our current methods is that evaluating UX, especially in large system designs, is very difficult.

No one has the resources to look everywhere and test every possible feature on every possible screen or web page in every possible task. You are just not going to find all the UX problems in all those places. One evaluator might find a problem in a place that other evaluators didn't even look. Why are we surprised that each evaluator doesn't come up with the same comprehensive problem list?

How can you ever hope to find your way through it all, let alone do a thorough job of UX evaluation? There are just so many issues and difficulties, so many places for UX problems to hide. It brings to mind the image of a person with a metal detector, searching over a large beach. There is no chance of finding all the detectable items.

28.3.3 “Discount” UX Evaluation Methods

28.3.3.1 What is a “discount” UX evaluation method?

A rapid UX evaluation method is a method that results simply from trading off the thoroughness of high rigor for low cost and fast application. The rapid methods include most of the analytic data collection, especially in their less rigorous forms. Because inspection techniques are less costly, they have been called “discount” evaluation techniques (Nielsen, 1989).

Inspection (UX)

An analytical evaluation method in which a UX expert evaluates an interaction design by looking at it or trying it out, sometimes in the context of a set of abstracted design guidelines. Expert evaluators are both participant surrogates and observers, asking themselves questions about what would cause users problems and giving an expert opinion predicting UX problems (Section 25.4).

Subject Matter Expert (SME)

Someone with a deep understanding of a specific work domain and the range of work practices within that domain (Section 7.4.4.1).

28.3.3.2 Criticism of discount methods

Rapid UX evaluation methods, especially inspection methods, have been criticized as “damaged merchandise” (Gray and Salzman, 1998) or “discount goods” (Cockton and Woolrych, 2002). The term “rapid” was intended as a positive reflection of the advantage of lower costs. However, the terms “damaged” and “discount” were used pejoratively to connote inferior, bargain-basement goods because of the reduced effectiveness and susceptibility to errors in identifying UX problems.

28.3.3.3 Real limitations

The major downsides of inspection methods, which do not employ real users, is that they can be error-prone and can tend to find a higher proportion of lower severity problems. They can suffer from validity problems, yielding some false positives (occurrences identified as UX issues but which turn out not to be real problems) and missing some UX problems because of false negatives.

Another potential drawback is that the UX experts doing the inspection may not know the subject-matter domain or the system in depth. This can lead to a less effective inspection but can be offset somewhat by including a subject-matter expert on the inspection team.

It might be some comfort, however, to know that these weaknesses are inherent in virtually all kinds of UX evaluation, including our venerable yardstick of performance, rigorous lab-based empirical formative evaluation (Molich et al., 1998; Molich et al., 1999; Newman, 1998; Spool and Schroeder, 2001). That is because many of the phenomena and principles are the same, and the working concepts are not that different. Formative evaluation, in general, just isn’t very reliable or repeatable.

28.3.3.4 But do less rigorous methods work?

As in most things, the value of these methods depends on the context of their use and the most common use today is in agile UX practice, where the rapid evaluation methods are the de facto standard.

In the hands of an experienced evaluator, inspection methods can be relatively fast and very effective—you can get a significant portion of UX problems dealt with and out of the way at a low cost. Under the right conditions, you can do a UX inspection and its analysis, fix the major problems, and update the prototype design in a day!

28.3.3.5 *Be practical*

Approaching perfection is expensive. In engineering, the goal is to make it good enough. Wixon (2003) speaks up for the practitioner in the discussion of usability evaluation methods. From an applied perspective, he points out that a focus on validity and proper statistical analysis in these studies isn't serving the needs of practitioners in finding the most suitable usability evaluation method and best practices for their work context in the business world.

Wixon (2003) would like to see more emphasis on usability evaluation method comparison criteria that take into account factors that determine their success in real product development organizations. As an example, the value of a method might be less about absolute numbers of problems detected and more about how well a usability evaluation method fits into the work style and development process of the organization. And to this we can only say amen.

28.3.3.6 *Sometimes you do have to pay more to get more*

Where human lives are at risk, we are compelled to spend more resources to be more thorough in all our process activities and surely don't want an evaluation error to weaken the design in the end. In most applications, though, each error simply means that we missed an opportunity to improve usability in just one detail of the design.

28.3.3.7 *At the end of the day, discount methods are the way forward*

In sum, although criticized as false economy in the academic HCI literature, these so-called "discount" methods are practiced heavily and successfully in the field.

Hope. Among reasons we have to be optimistic in the long run about less rigorous UX evaluation methods are:

- Iteration helps close the gap.
- Evaluation methods can be backed up with UX expertise.
- Because evaluators have individual differences, adding evaluators really helps because one evaluator might detect a problem that others don't.
- We are using our experience as UX practitioners and not doing the process with our eyes closed.

Low reliability is acceptable. It's not that the facts uncovered in the studies critical of evaluation reliability are wrong but, in the context of UX practice, the conclusions are wrong. Despite the low reliability of methods, especially the rapid and low-cost methods; these methods work quite well enough to suffice. While our methods may not find all UX problems, we usually get some good ones. If we fix those, maybe we will get the others the next time.

So don't worry too much about evaluation reliability. Every problem you find and fix is one less barrier to a good user experience. It's an engineering process and we have to be happy with "good enough."

So, here we wholeheartedly affirm the value of discount UX evaluation methods among your UX engineering tools!

28.4 HISTORICAL ROOTS FOR UX METRICS AND TARGETS

The concept of formal UX measurement specifications in tabular form, with various metrics operationally defining success, was originally developed by [Gilb \(1987\)](#). The focus of Gilb's work was on using measurements in managing software development resources. [Bennett \(1984\)](#) adapted this approach to usability specifications as a technique for setting planned usability levels and managing the process to meet those levels.

These ideas were integrated into usability engineering practice by [Good, Spine, Whiteside, and George \(1986\)](#) and further refined by [Whiteside, Bennett, and Holtzblatt \(1988\)](#). Usability engineering, as defined by [Good et al. \(1986\)](#), is a process through which quantitative usability characteristics are specified early and measured throughout the lifecycle process.

28.5 THE EARLY A330 AIRBUS—AN EXAMPLE OF THE NEED FOR ECOLOGICAL VALIDITY IN TESTING

We experienced a real-world example of a product that could have benefited enormously from better ecological validity (the extent to which your UX evaluation setup matches the user's real work context) in its testing. We traveled in an A330 Airbus airplane when that model first came out; our plane was 1 week old. (Advice: for many reasons, do not be the first to travel in a new airplane design.) We were told that a human-centered approach was taken in the A330 Airbus design, including UX testing of buttons, layout, and so on of the passenger

controls for the entertainment system. Apparently, though, they did not do enough in situ testing. Each passenger had a personal viewing screen for movies and other entertainment, considered an advantage over the screens hanging from the ceiling. The controls for each seat were more or less like a TV remote control, only tethered with a “pull-out” cord. When not in use, the remote control snapped into a recessed space on the seat arm rest. Cool, eh?

The LCD screen had nice color and brightness but a low acceptable viewing angle. Get far off the axis (away from perpendicular to the screen) and you lose all brightness, and, just before it disappears altogether, you see the picture as a color negative image. But the screen is right in front of you, so no problem, right? Right, until in a real flight, the person in front of you tilts back the seat. Then, we could barely see it. We could tell it was affecting others, too, because we could see many people leaning their heads down into an awkward position just to see the screen. After a period of fatigue, many people gave up, turned it off, and leaned back for comfort. If the display screen was used in UX testing, and we have to assume it was, the question of tilting the seat never entered the discussion, probably because the screen was propped up on a stand in front of each participant in the UX lab. Designers and evaluators just did not think about passengers in front of screen users tilting back their seats. Testing in a more realistic setting, better emulating the ecological conditions of real flight, would have revealed this major flaw.

It does not end there. Once the movie got going, most people stowed the remote control away in the arm rest. But, of course, what do you also rest on an arm rest? Your arm. And in so doing, it was easy to bump a button on the control and make some change in the “programming.” The design of this clever feature almost always made the movie disappear at a crucial point in the plot. And because we were syncing our movie viewing, the other one of us had to pause the movie, while the first one had to go back through far too many button pushes to get the movie back and fast-forwarded to the current place.

It still does not end here. After the movie was over (or for some viewers, after they gave up) and we wanted to sleep, a bump of the arm on the remote caused the screen to light up brightly, instantly waking us to the wonderful world of entertainment. The flight attendant in just one week with this design had already come up with a creative workaround. She showed us how to pull the remote out on its cord and dangle it down out of the way of the arm rest. Soon, and this is the UX-gospel truth, almost everyone in the plane had a dangling remote control swinging gracefully in the aisle like so many synchronized reeds in the breeze as the plane moved about on its course.

Participant

A participant, or user participant, is a user, potential, or user surrogate who helps evaluate UX designs for usability and user experience. These are the people who perform tasks and give feedback while we observe and measure. Because we wish to invite these volunteers to join our team and help us evaluate designs (i.e., we want them to participate), we use the term “participant” instead of “subject” (Section 21.1.3).

28.6 DETERMINING THE RIGHT NUMBER OF PARTICIPANTS

One of your activities in preparing for formative evaluation is finding appropriate users for the evaluation sessions. In formal summative evaluation, this part of the process is referred to as “sampling,” but that term isn’t appropriate here because what we are doing has nothing to do with the implied statistical relationships and constraints.

28.6.1 How Many are Needed? A Difficult Question

How many participants are enough? This is one of those issues that some novice UX practitioners take so seriously, and yet, it is a question to which there is no definitive answer. Indeed, there can’t be one answer. It depends so much on the specific context and parameters of your individual situation that you have to answer this question for yourself each time you do formative evaluation.

There are studies that lead UX gurus to proclaim various rules of thumb, such as, “three to five users are enough to find 80% of your UX problems,” but when you see how many different assumptions are used to arrive at those “rules,” and how few of those assumptions are valid within your project, you realize that this is one place in the process where it is most important for you to use your own head and not follow vague generalizations.

And, of course, cost is often a limiting factor. Sometimes, you just get one or two participants in each of one or two iterations, and you have to be satisfied with that because it is all you can afford. The good news is that you can accomplish a lot with only a few good participants. There is no statistical requirement for large numbers of “subjects” as there is for formal summative evaluation; rather, the goal is to focus on extracting as much information as possible from every participant.

28.6.2 Rules of Thumb Abound

There are bona fide studies that predict the optimum number of participants needed for UX testing under various conditions. Most “rules of thumb” are based empirically but, because they are quoted and applied so broadly without regard to the constraints and conditions under which the results were obtained, these rules have become among the most folklorish of folklore out there.

Nielsen and Molich (1990) had an early paper about the number of users/participants needed to find enough UX problems and found that 80% of their known UX problems could be detected with four to five participants, and the

most severe problems were usually found with the first few participants. Virzi (1990, 1992) more or less confirmed Nielsen and Molich's study.

Nielsen and Landauer (1993) found that detection of problems as a function of the number of participants is well modeled as a Poisson process, supporting the ability to use early results to estimate the number of problems left to be found and the number of additional participants needed to find a certain percentage.

Depending on the circumstances, though, some say that even five participants is no way near enough (Hudson, 2001; Spool and Schroeder, 2001), especially for complex applications or large websites. In practice, each of these numbers has proven to be right for some set of conditions, but the question is whether they will work for you in your evaluation.

28.6.3 An Analytic Basis for the Three-To-Five-Users Rule

28.6.3.1 The underlying probability function

In Fig. 28-1, you can see graphs, related to the binomial probability distribution, of cumulative percentages of problems likely to be found for a given number of participants used and at various detection rates, adapted from Lewis (1994).

Y-axis values in these curves are for “discovery likelihood,” expressed as a cumulative percentage of problems likely to be found as a function of the

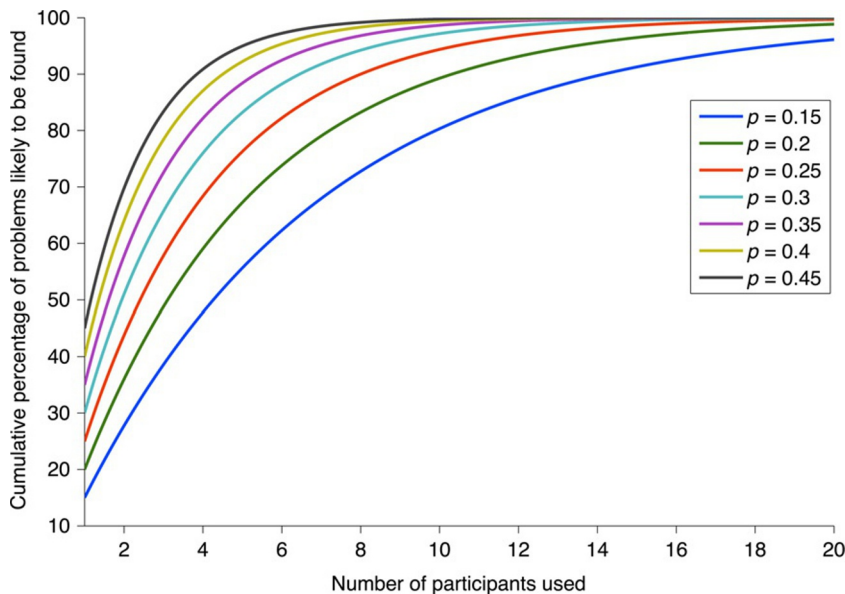


Fig. 28-1

Graphs of cumulative percentages of problems likely to be found for a given number of participants used and at various detection rates [adapted from (Lewis, 1994)].

number of participants or evaluators used. These curves are based on the probability formula:

discovery likelihood (cumulative percentage of problems likely to be found = $1 - (1 - p)^n$, where n is the number of participants used (X-axis values), and p is what we call the “detection rate” of a certain category of participants.

As an example, this formula tells us that a sample size of five participant evaluators (n) with an individual detection rate (p) of at least 0.30 is sufficient to find approximately 80% of the UX problems in a system.

28.6.3.2 The old balls-in-an-urn analogy

Let us think of an interaction design containing flaws that cause UX problems as analogous to the old probability setting of an urn containing various colored balls. Among an unknown number of balls of all colors, suppose there are a number of red balls, each representing a different UX problem.

Suppose now that a participant or evaluator reaches in and grabs a big handful of balls from the urn. This is analogous to an evaluation session using a single expert evaluator, if it is a UX inspection evaluation, or a single participant, if it is a lab-based empirical session. The number of red balls in that handful is the number of UX problems identified in the session.

In a UX inspection, it is the expert evaluator, or inspector, who finds the UX problems. In an empirical UX test, participants are a catalyst for UX problem detection—not necessarily detecting problems themselves, but encountering critical incidents while performing tasks, enabling evaluators to identify the corresponding UX problems. Because the effect is essentially the same, for simplicity in this discussion we will use the term “participant” for both the inspector and the testing participant and “find problems” for whatever way the problems are found in a session.

28.6.3.3 Participant detection rates

The detection rate, p , of an individual participant is the percentage of existing problems that this participant can find in one session. This corresponds to the number of red balls a participant gets in one handful of balls. This is a function of the individual participant. For example, in the case of the balls in the urn, it might be related to the size of the participant’s hand. In the UX domain, it is perhaps related to the participant’s evaluation skills.

In any case, in this analysis, if a participant has a detection rate of $p = 0.20$, it means that this participant will find 20% of the UX problems existing in the design. The number of participants with that same individual detection rate who,

Critical Incident

An event that occurs during user task performance or other user interaction that indicate possible UX problem(s). Critical incident identification, an empirical UX evaluation data collection technique based on the participant and/or evaluator detecting and analyzing critical incidents, is arguably the single most important qualitative data collection technique (Section 24.2.1).

in turn, reach into the urn is the value on the X axis. The curve shown with a green line is for a detection rate of $p = 0.20$. The other curves are for different detection rates, from $p = 0.15$ up to $p = 0.45$.

Most of the time, we do not even know the detections rates of our participants. To calculate the detection rate for a participant, we would have to know how many total UX problems exist in a design. But that is just what we are trying to find out with evaluation. You could, we guess, run a testing session with the participant against a design with a certain number of known flaws. But that would tell you that participant's detection rate for that day, in that context, and for that system. Unfortunately, a given participant's detection rate isn't constant.

28.6.3.4 Cumulative percentage of problems to be found

The Y axis represents values of the cumulative percentage of problems to be found. Let us look at this first for just one participant. The curve for $p = 0.20$, for example, has a Y axis value of 20%, for $n = 1$ (where the curve intersects the Y axis). This is consistent with our expectation that one participant with $p = 0.20$ will find 20% of the problems, or get 20% of the red balls, in the first session.

Now, what about the “cumulative” aspect? What happens when the second participant reaches into the urn depends on whether you replaced the balls from the first participant. This analysis is for the case where each participant returns all the balls to the urn after each “session”; that is, none of the UX problems are fixed between participants.

After the first participant has found some problems, there are fewer *new* problems left to find by the second participant. If you look at the results with the two participants independently, they each help you find a somewhat different 20% of the problems, but there is likely to be overlap, which reduces the cumulative effect (the union of the sets of problems) of the two.

This is what we see in the curves of Fig. 28-1 as the percentage of problems likely to be found drops off with each new participant (moving to the right on the X axis) because the marginal number of new problems found is decreasing. That accounts for the leveling off of the curves until, at some high number of participants, essentially no new problems are being found, and the curve is asymptotically flat.

28.6.3.5 Marginal added detection and cost-benefit

One thing we do notice in the curves of Fig. 28-1 is that, despite the drop-off of effective detection rates, as you continue to add more participants, you will continue to uncover more problems. At least for a while. Eventually, high detection rates coupled with high numbers of participants will yield results

that asymptotically approach about 100% in the upper right-hand part of the figure, and virtually no new problems will be found with subsequent participants.

But what happens along the way? Each new participant helps you find fewer new problems, but because the cost to run each participant is about the same, with each successive participant, the process becomes less efficient (fewer new problems found for the same cost).

As a pretty good approximation of the cost to run a UX testing session with n participants, you have a fixed cost to set up the session plus a variable cost (or cost per participant) = $a + bn$. The benefit of running a UX testing session with n participants is the discovery likelihood. So, the cost benefit is the ratio benefit/cost, each as a function of n , or $\text{benefit/cost} = (1 - (1 - p)^n) / (a + bn)$.

If you graph this function (with some specific values of a and b) against $n = 1, 2, \dots$, you will see a curve that climbs for the first few values of n and then starts dropping off. The values of n around the peak of cost-benefit are the optimum (from a cost-benefit perspective) number of participants to use. The range of n for which the peak occurs depends on parameters a , b , and p of your setup; your mileage can vary.

Nielsen and Landauer (1993) showed that real data for both UX inspections and lab-based testing with participants did match this mathematical cost-benefit model. Their results showed that, for their parameters, the peak occurred for values of n around 3 to 5. Thus, the “three-to-five-users” rule of thumb.

28.6.3.6 Assumptions do not always apply in the real world

This three-to-five-users rule, with its tidy mathematical underpinning, can and does apply to many situations similar to the conditions Nielsen and Landauer (1993) used, and we believe their analysis brings insight into the discussion. However, we know there are many cases where it just does not apply.

For starters, all of this analysis, including the analogy with the balls-in-an-urn setting, depends on two assumptions:

- Each participant has a constant detection rate, p .
- Each UX problem is equally likely to be found in testing.

If UX problems were balls in an urn, our lives would be simpler. But neither of these assumptions is true, and the UX life isn’t simple.

Assumptions about detection rates. Each curve in Fig. 28-1 is for a fixed detection rate, and the cost-benefit calculation given earlier was based on a fixed detection rate, p . But the “evaluator effect” tells us not only will different evaluators find

different problems, but also the detection rate can vary widely over participants (Hertzum and Jacobsen, 2003).

In fact, a given individual does not even have a fixed “individual detection rate”; it can be influenced from day to day or even from moment to moment by how rested the participant is, blood caffeine and ethanol levels, attitude, the system, how the evaluators conduct the evaluation, what tasks are used, the evaluator’s skills, and so on.

Also, what does it *really* mean for a testing participant to have a detection rate of $p = 0.20$? How long does it take in a session for that participant to achieve that 20% discovery? How many tasks? What kinds of tasks? What if that participant continues to perform more tasks? Will no more critical incidents be encountered after 20% detection is achieved?

Assumptions about problem detectability. The curves in Fig. 28-1 are also based on an assumption that all problems are equally detectable (like all red balls in the urn are equally likely to be drawn out). But, of course, we know that some problems are almost obvious on the surface, and other problems can be orders of magnitude more difficult to ferret out. So detectability, or likelihood of being found, can vary dramatically across various UX problems.

Task selection. One reason for the overlap in problems detected from one participant to another, causing the cumulative detection likelihood to fall off with additional participants, as it does in Fig. 28-1, is the use of prescribed tasks. Participants performing essentially the same sets of tasks are looking in the same places for problems and are, therefore, more likely to uncover many of the same problems.

However, if you employ user-directed tasks (Spool and Schroeder, 2001), participants will be looking in different places, and the overlap of problems found could be much less. This keeps the benefit part of the curves growing linearly for more participants, causing your optimum number of participants to be larger.

Application system effects. Another factor that can torpedo the three-to-five-users rule is the application system being evaluated. Some systems are much larger than others. For example, an enormous website or a large and complex word processor will harbor many more possibilities for UX problems than, say, a simple interoffice scheduling system. If each participant can explore only a small portion of such an application, the overlap of problems among participants may be insignificant. In such cases, the cost-benefit function will peak with many more participants than three to five.

Bottom line. You can’t compute and graph all the theoretical curves and parameters we talk about here because you will never know how many UX

problems exist in the design, so you will never know what percentage of the existing problems you have found. You do not have to use those curves, anyway, to have a pretty good intuitive feeling for whether you are still detecting useful new problems with each new participant. Look at the results from each participant and each iteration and ask if those results were worth it and whether it is worth investing a little more.

Critical Incident

An event that occurs during user task performance or other user interaction that indicate possible UX problem(s). Critical incident identification, an empirical UX evaluation data collection technique based on the participant and/or evaluator detecting and analyzing critical incidents, is arguably the single most important qualitative data collection technique (Section 24.2.1).

28.7 HISTORICAL ROOTS OF THE CRITICAL INCIDENT TECHNIQUE

28.7.1 Critical Incident Techniques Started Long Ago in Human Factors

The origins of the critical incident technique can be traced back at least to studies performed in the Aviation Psychology Program of the US Army Air Forces in World War II to analyze and classify pilot error experiences in reading and interpreting aircraft instruments. The technique was first formally codified by the work of [Fitts and Jones \(1947\)](#). [Flanagan \(1954\)](#) synthesized the landmark critical incident technique.

28.7.2 Mostly Used as a Variation

When Flanagan designed the critical incident technique in 1954, he did not see it as a single rigid procedure. He was in favor of modifying this technique to meet different needs as long as original criteria were met. The variation occurring over the years, however, may have been more than Flanagan anticipated. Forty years after the introduction of Flanagan's critical incident technique, [Shattuck and Woods \(1994\)](#) reported a study that revealed that this technique has rarely been used as originally published. In fact, numerous variations of the method were found, each suited to a particular field of interest. In HCI, we have continued this tradition of adaptation by using our own version of the critical incident technique as a primary UX evaluation technique to identify UX problems and their causes.

28.7.3 Who Identifies Critical Incidents?

In the original work by [Fitts and Jones \(1947\)](#), the user (an airplane pilot) was the one who reported critical incidents after task performance was completed. Later, [Flanagan \(1954\)](#) used trained observers (evaluators) to collect critical incident information while observing users performing tasks.

del Galdo, Williges, Williges, and Wixon (1986) involved users in identifying their own critical incidents, reporting during task performance. The technique was also used as a self-reporting mechanism by Hartson and Castillo (1998) as the basis for remote system or product usability evaluation. Further, Dzida, Wiethoff, and Arnold (1993) and Koenemann-Belliveau, Carroll, Rosson, and Singley (1994) adopted the stance that identifying critical incidents during task performance can be an individual process by either the user or an evaluator or a mutual process between the user and an evaluator.

28.7.4 Timing of Critical Incident Data Capture: The Evaluator's Awareness Zone

While users are known to report major UX problems in alpha and beta testing (sending software out for comments on how well it worked), one reason these methods can't be relied upon for thorough identification of UX problems to fix is the retrospective nature of that kind of data collection. Lab-based UX evaluation has the advantage of having the precious and volatile details right in front of you as they happen. The key to this kind of UX data is in the details, and details of these data are perishable; they must be captured immediately as they arise during usage.

The upshot: capture and document the details while they are fresh. If you capture them as they happen, we call it concurrent data capture. If you capture data immediately after the task, we call it contemporaneous data capture. If you try to capture data after the task is well over, through someone trying to remember the details in an interview or survey after the session, this is retrospective data capture, and many of the once-fresh details can be lost.

It isn't as easy, however, as just capturing critical incident data immediately upon its occurrence. A critical incident is often not immediately recognized for what it is. In [Fig. 28-2](#), the evaluator's recognition of a critical incident will necessarily occur sometime after it begins to occur. And following the point of initial awareness, after confirming that it is a critical incident, the evaluator requires some time and thought in a kind of "awareness zone" to develop an understanding of the problem, possibly through discussion with the participant.

The optimum time to report the problem, the time when the potential for a quality problem report is highest, is at the peak of this problem understanding, as seen in [Fig. 28-2](#). Before that point, the evaluator has not yet established a full understanding of the problem. After that optimum point, natural abstraction due to human memory limitations sets in and

Alpha and Beta Testing

Post-deployment evaluation methods in which near-final versions of a product are sent to select users, experts, customers, and professional reviewers as a preview, in exchange for which, they try it out and give feedback on the experience ([Section 21.5.1](#)).

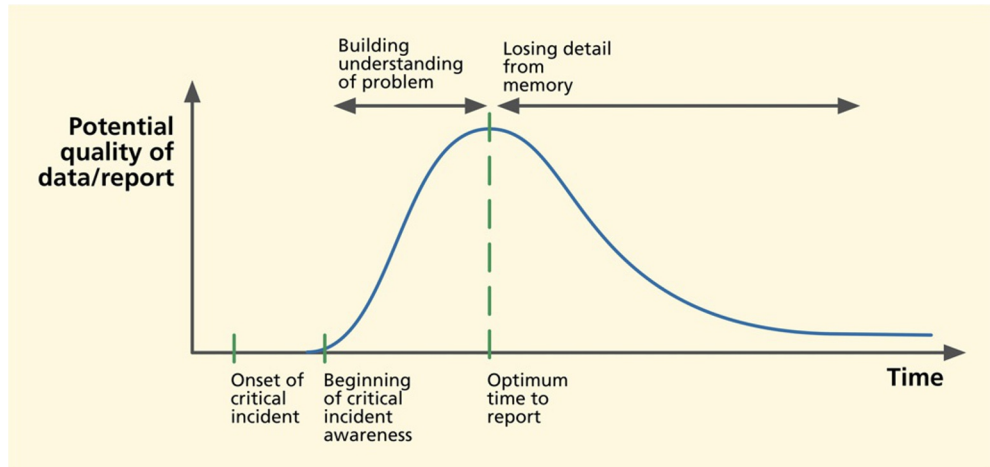


Fig. 28-2

*Critical incident
description detail versus
time after critical incident.*

details drop off rapidly with time, accelerated by proactive interference from any intervening tasks.

28.8 OTHER METHODS FOR IDENTIFYING EMOTIONAL RESPONSE TO UX DESIGNS

28.8.1 Direct Observation of Physiological Responses as Indicators of Emotional Impact

In contrast to self-reporting techniques, UX practitioners can obtain emotional impact indicator data through direct observation of participant physiological responses to emotional impact encounters as usage occurs. Usage can be teeming with user behaviors, including facial expressions (such as ephemeral grimaces or smiles) and body language (such as tapping of fingers, fidgeting, or scratching one's head) that indicate emotional impact. You will need to create conditions to encourage episodes of long-term usage activity to occur during these observational periods.

Physiological responses can be “captured” either by direct behavioral observation or by instrumented measurements. Behavioral observations include those of facial expressions, gestural behavior, and body posture. One difficulty in any observation or measurement of physiological reactions to usage events is that often there is no way to connect the physiological response to a particular emotion and its cause within the interaction.

The emotional “tells” of facial and bodily expressions can be fleeting and subliminal, easily missed in real-time observation. Therefore, to capture facial expressions data and other similar observational data reliably, practitioners usually need to make video recordings of participant usage behavior and do frame-by-frame analysis. Methods for interpreting facial expressions have been developed, including one called the Facial Action Coding System (Ekman and Friesen, 1975).

Kim et al. (2008) remind us that while we can measure physiological effects, it is difficult to connect the measurements with specific emotions and with causes within the interaction. Their solution is to supplement with traditional synchronized video-recording techniques to correlate measurements with usage occurrences and behavioral events. But this kind of video review has disadvantages: the reviewing process is usually tedious and time-consuming, you may need an analyst trained in identifying and interpreting these expressions often within a frame-by-frame analysis, and even a trained analyst can’t always make the right call.

Fortunately, software-assisted recognition of facial expressions and gestures in video images is beginning to be feasible for practical applications. Software tools are now becoming available to automate real-time recognition and interpretation of facial expressions. A system called “faceAPI”¹ from Seeing Machines is advertised to both track and understand faces. It comes as a software module that you embed in your own product or application. An ordinary webcam, focused on the user’s face, feeds both faceAPI and any digital video-recording program with software-accessible time stamps and/or frame numbers.

Facial expressions do seem to be mostly culture independent, and you can capture expressions without interruption of the usage. However, there are limitations that generally preclude their use. The main limitation is that they are useful for only a limited set of basic emotions such as anger or happiness, but not mixed emotions. Dormann (2003) says it is, therefore, difficult to be precise about what kind of emotion is being observed.

In order to identify facial expressions, faceAPI must track the user’s face during head movement that occurs in 3D with usage. The head-tracking feature outputs X, Y, Z position and head orientation coordinates for every video frame. The facial feature detection component of faceAPI tracks three points on each eyebrow and eight points around the lips.

The detection algorithm is “robust to occlusions, fast movements, large head rotations, lighting, facial deformation, skin color, beards, and glasses.” This part

¹<http://www.seeingmachines.com/product/faceapi/>.

of faceAPI outputs a real-time stream of facial feature data, time coordinated with the recorded video, that can be understood and interpreted via a suite of image-processing modules. The faceAPI system is a commercial product, but a free version is available to qualified users for noncommercial use.

28.8.2 Biometrics to Detect Physiological Responses to Emotional Impact

The use of instrumented measurement of physiological responses in participants is called biometrics. Biometrics are about detection and measurement of autonomic or involuntary bodily changes triggered by nervous system responses to emotional impact within interaction events. Examples include changes in heart rate, respiration, perspiration, and eye pupil dilation. Changes in perspiration are measured by galvanic skin response measurements to detect changes in electrical conductivity.

Such nervous system changes can be correlated with emotional responses to interaction events. Pupillary dilation is an autonomous indication especially of interest, engagement, and excitement and is known to correlate with a number of emotional states (Tullis and Albert, 2008).

The downside of biometrics is the need for specialized monitoring equipment. If you can get some good measuring instruments and are trained to use them to get good measures, it does not get more “embodied” than this. But most equipment for measuring physiological changes is out of reach for the average UX practitioner.

It is possible to adapt a polygraph or lie detector, for example, to detect changes in pulse, respiration, and skin conductivity that could be correlated with emotional responses to interaction events. However, the operation of most of this equipment requires skills and experience in medical technology, and interpretation of raw data can require specialized training in psychology, all beyond our scope. Finally, the extent of culture independence of facial expressions and other physiological responses isn’t entirely known.

28.8.3 The HUMAINE Project—Physiological Techniques for Affective Measurements

The European community project HUMAINE (Human-Machine Interaction Network on Emotions) issued a technical report detailing a taxonomy of affective measurement techniques (Westerman, Gardner, and Sutherland, 2006). They point out that there is a history of physiological and psychophysiological measurement in human factors practice since the late 1970s to detect, for

Embodied Interaction

Interaction with technology that involves a user’s body in a natural and significant way, such as by using gestures (Section 6.2.6.3).

example, stress due to operator overload, and an even longer history of this kind of measurement in psychological research.

In the HUMAINE report, the authors discuss the role of medicine in physiological measurement, including electroencephalograms and event-related potential, measured with electroencephalography, a technique that detects and measures electrical activity of the brain through the skull and scalp. Event-related potentials can be roughly correlated to cognitive functions involving memory and attention and changes in mental state.

As the authors say, these physiological measurements have the advantage over self-reporting methods in that they can monitor continuously, require no conscious user actions, and do not interrupt task performance or usage activity. To be meaningful, however, such physiological measurements have to be associated with time stamps on a video of user activity.

A major disadvantage, ruling the approach out for most routine UX evaluation, is the requirement for attached sensors. New, less intrusive instrumentation is being developed. For example, [Kapoor, Picard, and Ivanov \(2004\)](#) report being able to detect changes in user posture, for example, due to fidgeting, with pressure sensors attached to a chair.

28.9 NIELSEN AND MOLICH'S ORIGINAL HEURISTICS

The first set of heuristics that Nielsen and Molich developed for usability inspection ([Molich and Nielsen, 1990](#); [Nielsen and Molich, 1990](#)) were 10 “general principles” for interaction design. They called them heuristics because they are not strict design guidelines. Here, we list the 10 original Nielsen and Molich heuristics from Nielsen’s *Usability Engineering* book ([Nielsen, 1993, Chapter 5](#)).

- Simple and natural dialogue:
 - Good graphic design and use of color.
 - Screen layout by gestalt rules of human perception.
 - Less is more; avoid extraneous information.
- Speak the users’ language:
 - User-centered terminology, not system or technology centered.
 - Use words with standard meanings.
 - Vocabulary and meaning from work domain.
 - Use mappings and metaphors to support learning.

Heuristic (in UX Design)

An analytic evaluation method based on expert UX inspection guided by a set of heuristics, general high-level UX design guidelines ([Section 25.5](#)).

- Minimize user memory load:
 - Clear labeling.
- Consistency:
 - Help avoid errors, especially by novices.
- Feedback:
 - Make it clear when an error has occurred.
 - Show user progress.
- Clearly marked exits:
 - Provide escape from all dialogue boxes.
- Shortcuts:
 - Help expert users without penalizing novices.
- Good error messages:
 - Clear language, not obscure codes.
 - Be precise rather than vague or general.
 - Be constructive to help solve problem.
 - Be polite and not intimidating.
- Prevent errors:
 - Many potential error situations can be avoided in design.
 - Select from lists, where possible, instead of requiring user to type in.
 - Avoid modes.
- Help and documentation:
 - When users want to read the manual, they are usually desperate.
 - Be specific with online help.

Information Object

An internally stored structured or simple article, piece of information, or data that serves as a work object. Often data entities are central to work flow, being operated on by users; they are organized, shared, labeled, navigated, searched and browsed for, accessed and displayed, modified and manipulated, and stored back again, all within a system ecology (Section 14.2.6.7).

28.10 UX PROBLEM DATA MANAGEMENT

As time goes by, and you proceed further into the UX process lifecycle, the full life story of each UX problem grows, entailing slow expansion of data in the UX problem record. Each UX problem record will eventually contain information about the problem: diagnosis by problem type and subtype, interaction design flaws as problem causes, cost/importance data estimating severity, management decisions to fix (or not) the problem, costs, implementation efforts, and downstream effectiveness.

Most authors mention UX problems or problem reports but do not hint at the fact that a complete problem record can be a large and complex information object. Maintaining a complete record of this unit of UX data is surely one place where some kind of tool support, such as a database management system, is warranted. As an example of how your UX problem record structure and content

can grow, here are some of the kinds of information that can eventually be attached to it. These are possibilities we have encountered; pick the ones that suit you:

- Problem name.
- Problem description.
- Task context.
- Effects on users (symptoms).
- Associated designer knowledge.
- Problem diagnosis (problem type and subtype and causes within the design).
- Links to constituent UX problem instances.
- Links for relationships to other UX problems (e.g., in groups to be fixed together).
- Links to project context.
- Project name.
- Version/release number.
- Project personnel.
- Link to evaluation session.
- Evaluation session date, location, etc.
- Session type (e.g., lab-based testing, UX inspection, remote evaluation).
- Links to evaluators.
- Links to participants.
- Cost-importance attributes for this iteration (next section).
- Candidate solutions.
- Estimated cost to fix.
- Importance to fix.
- Priority ratio.
- Priority ranking.
- Resolution.
- Treatment history.
- Solution used.
- Dates, personnel involved in redesign, implementation.
- Actual cost to fix.
- Results (e.g., based on retesting).

For more about representation schemes for UX problem data, see [Lavery and Cockton \(1997\)](#).